

Muhammad Umar Khan

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EDUCATION

Wayne State University | Detroit, MI, USA

Jan. 2025 - Dec. 2026

M.S. in Computer Science · GPA: 3.89/4

University of Windsor | Windsor, ON, Canada

Sep. 2020 – Apr. 2024

B.S. in Computer Science – Software Engineering Specialization

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript/TypeScript, HTML, CSS, SQL

Frameworks & Libraries: Spring Boot, Spring Security, Hibernate/JPA, React, Next.js, Node.js, Vite, Tailwind CSS, Recharts, Flask, Ollama

Databases & Tools/DevOps: PostgreSQL, SQLite, MongoDB, Supabase, Git, GitHub, Docker, REST APIs, JWT

Data & AI/ML: NumPy, Pandas, Matplotlib, TensorFlow, PyTorch, Scikit-learn, CNN+LSTM, Time Series Forecasting, RAG, TF-IDF

PROJECTS

MindForge AI Suite (StudyAI + ResearchAI) | Python, Flask, Ollama, LLaMA 3.2, RAG, TF-IDF, ThreadPoolExecutor

- Built a local AI study and research suite powered by LLaMA 3.2 via Ollama, running 100% offline with support for PDF, DOCX, PPTX, and TXT formats.
- Implemented a RAG pipeline with TF-IDF chunk retrieval, reducing context window usage and improving response relevance across multi-document research sessions
- Optimized multi-document analysis throughput by ~75% using parallel summarization via ThreadPoolExecutor, with real-time SSE streaming for live progress updates
- Built cross-document synthesis engine with web search integration (DuckDuckGo), APA citation generation, and structured research report generation via LLM

Market Predictor | React, Next.js, Supabase, Claude AI, Recharts, JavaScript

- Implemented a custom CNN+LSTM neural network from scratch in JavaScript for stock & crypto price forecasting, training live over 30 epochs with real-time loss curve visualization across 1W/2W/1M horizons
- Integrated Claude AI for chart image analysis (OCR + pattern recognition), AI investment insights, and news sentiment — routed through a secure Next.js server-side API proxy
- Built a prediction accuracy tracker that resolves forecasts against real market prices post-horizon and benchmarks CNN+LSTM against flat and momentum baselines

Blog Platform (Full-Stack) | Java, Spring Boot, Spring Security, JWT, React, TypeScript, Vite, Tailwind CSS, PostgreSQL

- Built a secure full-stack blog platform with JWT authentication and role-based access control using Spring Boot and Spring Security, and architected RESTful APIs with layered architecture (Controller → Service → Repository).
- Developed a responsive React + TypeScript frontend with CRUD operations, dynamic filtering, secure password hashing, token refresh flow, and protected route handling for session management.

K-Means Market Basket Analysis | Java

- Implemented K-Means clustering and the Apriori algorithm from scratch on a ~25% subset of a 530k+ entry dataset to analyze customer purchasing patterns, while generating association rules and cluster insights for data-driven decision making.

EXPERIENCE

University of Windsor | Windsor, ON, Canada

Apr. 2023 - Nov. 2024

Research Assistant

- Engineered a Python-based image segmentation pipeline using Euclidean-distance clustering to generate super pixels, reducing preprocessing time by approximately 20%.
- Evaluated multiple CNN and Graph-CNN models for pixel classification, documented results, and recommended the best model based on accuracy and runtime performance.
- Modeled and integrated an SQL database with Snowflake and containerized prototypes using Docker.

University of Windsor | Windsor, ON, Canada

Sep. 2022 – Apr. 2024

Teaching Assistant

- Conducted weekly office hours and graded assignments/exams for 100+ students enrolled in Data Structures & Algorithms, Database Management Systems, and Object-Oriented Programming (Java).
- Assisted lab sessions and exams, and aided in reinforcing core concepts in algorithms, database systems, and OOP.